

		LAYER	HARDWARE	ADDRESSING	PROTOCOL	TERM	
Software	7	Application	Software applications (Browsers, Email Clients)	FQDN (Domain Names), URLs.	HTTP, HTTPS, FTP, SMTP, DNS, SSH, DHCP	Data, User Interface	
	6	Presentation	Redirectors, Gateways	Data syntax formats	SSL/TLS, ASCII, JPEG, GIF, MPEG.	Data Formatting, Encryption, Compression	
	5	Session	Application Servers, load balancers, gateways	Session ID, Logical ports	NetBIOS, RPC, SQL, PPTP (vpn)	Dialogue Control, Synchronization.	
	4	Transport	Host/Firewall, Firewalls, Gateways	Port numbers, TCP/UDP ports	TCP, UDP, SCTP, SPX	Segments (TCP) / Datagrams (UDP), Flow Control, Error Correction	
Hardware and Software	3	Network	Routers, layer 3 switches	Logical IP Address	IP- addressing/routing ICMP- error reporting ARP- map IP to MAC OSPF- internal routing BGP- external routing	Packet, Subnet mask, routing	
	2	Data Link	NIC, bridges, switches, Access Point	Physical MAC Address	Ethernet (802.3), Wi-Fi (802.11), (PPP), HDLC, Frame Relay, ARP, Switching/Bridging	Frame, Content Address Memory (CAM), Logical Link Control (LLC), Media Access Control (MAC) address	
	1	Physical	Hub	no addressing but information is sent as bits representing 1 and 0s	Ethernet (physical), DSL, ISDN, 802.11 (Wi-Fi)	data in flight, bits on the wire, bits on the wire, electrical signals, light pulses, radio waves,	
			cable, repeater, fiber, signal, NIC, media converter, transceiver				